

Predicting the Zero-Shot Transfer of a Promptable Video Tracker

Forecasting SAM 3's per-species, per-place reliability —
for detection and association — from label-free distances

Kostiantyn Boiar
Student ID: <TODO>

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School of Computing Science
University of Glasgow

Supervisor: Dr Marwa Mahmoud

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Abstract

Promptable trackers such as SAM3 segment and track any named animal in a video with no training on that species, yet no rule tells a practitioner *in advance* whether the model is reliable for a given animal in a given place. This dissertation asks whether such a rule exists. On the SA-FARI benchmark, whose splits are disjoint by species and location, we measure SAM3’s zero-shot promptable-tracking accuracy with the official evaluator, decompose it into detection (pDetA) and association (pAssA), and test whether each is predicted, *before running the model* and without any target label, by four distances (taxonomic, visual, environment, temporal). Under a deliberately powerful test — whole-species and whole-location hold-out with group-cluster-bootstrap significance — **it is not**: no distance beats a mean-predictor baseline out of sample, and the one apparent effect is an animal-size confound. A follow-up pivot to intrinsic scene difficulty (low-light, clutter) collapses to the same size effect, and the decomposition shows association carries almost no signal distinct from detection. The contribution is thus a rigorous, decomposed *negative* result — the first for a promptable video tracker — validated by a test shown to detect the one real signal (object size) when it is present.

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Chapter 1

Introduction

1.1 Motivation

Promptable video trackers such as SAM3 [4] segment and track any named animal in a video with no training on that species, and conservation teams now deploy them on new species and regions constantly. Yet no rule tells a practitioner, *before running the model*, whether its output is trustworthy for a given animal in a given place.

1.2 Problem statement and research question

This dissertation asks whether SAM3’s zero-shot tracking accuracy on an unseen (species, place) can be *predicted in advance* from properties measurable without any label of the target, and which factors govern that transfer — separately for *finding* the animal and *following* it.

1.3 Hypotheses

H0 (null)

the label-free distances carry no predictive information about pDetA/pAssA — transfer is not distance-driven.

H1 (detection)

pDetA falls with species novelty (taxonomic + visual distance).

H2 (association)

pAssA falls with environment difficulty (scene, day/night/IR, clutter, camera motion).

Failing to reject H0 is itself an informative result — and, as Chapter 4 shows, it is what we find: neither a novelty distance nor a follow-up *intrinsic-difficulty* pivot predicts transfer beyond a trivial animal-size effect. Representational probing (does SAM3 encode taxonomy?) remains the open branch for future work.

1.4 Contributions

To our knowledge this is the first study to *predict*, before the model is run and from label-free distances, the zero-shot transfer of a promptable video tracker — and the first to split that prediction into **detection** (pDetA) and **association** (pAssA). This distinguishes it from label-free performance prediction for classifiers, which estimates a single scalar from the model’s own outputs on the target [1, 6, 10], and from the very recent static-image detector/segmenter estimators [14, 16]; and from camera-trap shift studies, which measure but do not forecast per-site reliability [2, 3, 7]. Concretely we deliver: (i) a rigorous, out-of-sample *negative* result —

under a deliberately powerful test (whole-species and whole-location hold-out with group-cluster-bootstrap significance), label-free before-running distances do *not* predict SAM3’s transfer; (ii) the detection-versus-association decomposition, tested on two near-orthogonal splits — a constructed species hold-out (novelty) and the official location hold-out (environment); and (iii) a *hardened* account of *why* it fails: a follow-up intrinsic-difficulty pivot collapses to the same animal-size confound, so the only label-free property that predicts detection is object size — the very nuisance the analysis controls for — and that positive control confirms the test detects signal when it exists.

1.5 Dissertation outline

Chapter 2 reviews the related work; Chapter 3 describes the data, measurement, distance features, and modelling; Chapter 4 reports the results and out-of-sample validation; Chapter 5 discusses them; Chapter 6 concludes.

Chapter 2

Background and Related Work

This chapter places the dissertation among four bodies of work: promptable segmentation and tracking (the model we study), the tracking metric and its detection/association decomposition (how we score it), label-free prediction of out-of-distribution performance (the statistical idea we extend), and distribution shift in camera-trap vision (the application and the closest empirical neighbours). It closes by stating precisely the gap this work fills.

2.1 Promptable segmentation and tracking

The *Segment Anything* line moved segmentation from a fixed label set to an interactive, promptable formulation, and its successors extended prompting from points and boxes to *concepts*. SAM3 [4] is a promptable model that, given a short text prompt such as "impala", detects, segments, and tracks every matching instance across a video with no training on that category — a *zero-shot*, open-vocabulary detect–segment–track capability. It is exactly this capability that conservation teams now deploy on new species and regions, and it is the (frozen) model whose transfer we predict. SAM3 was released alongside SA-FARI [15], a large multi-animal camera-trap tracking dataset that we use as the substrate for our natural experiment (Section 2.6). We treat SAM3 strictly as a fixed black box: we never fine-tune it, and the only quantity we fit is a small regression *about* its behaviour.

2.2 The tracking metric and its decomposition

Higher Order Tracking Accuracy (HOTA) [9] evaluates multi-object tracking by combining, over a range of localisation thresholds, a *detection* term (**DetA**: were the right objects found?) and an *association* term (**AssA**: were identities kept consistent over time?). SA-FARI scores promptable tracking with a promptable variant, **pHOTA**, that decomposes identically into **pDetA** and **pAssA**. This decomposition is the intellectual core of the dissertation: rather than predict a single scalar, we ask whether *finding* and *following* an animal transfer for *different reasons* — the former with how novel the animal is, the latter with how difficult the scene is — and we therefore model the two components separately throughout.

2.3 Predictable out-of-distribution generalisation

A now-mature literature shows that a model’s performance under distribution shift can be anticipated *without* target labels. Miller et al. [10] observe that out-of-distribution (OOD) accuracy is strongly linearly correlated with in-distribution accuracy across many models and shifts, and Baek et al. [1] show that the *agreement* between a pair of models tracks OOD accuracy closely enough to estimate it from unlabelled data alone. A parallel strand estimates accuracy from a

model’s own statistics on the unlabelled target: average thresholded confidence [6] calibrates a confidence threshold on source data and reads off target accuracy as the mass above it, while Deng and Zheng [5] regress classifier accuracy directly on a *distribution-shift feature* — the Fréchet distance between train and test feature sets — which is the closest methodological cousin to our own “regress the score on a distance” approach.

Two properties characterise essentially all of this work, and both are limitations we depart from. First, it targets **image classifiers** with a single scalar accuracy. The idea has only recently reached structured outputs: Yoo et al. [16] estimate object-detection performance without ground truth from the spatial consistency and confidence reliability of pre-NMS boxes, and reverse classification accuracy [14] predicts per-image segmentation quality by training a reverse model against a labelled reference. These are the closest structured-output predecessors, but both operate on *static single images*, neither concerns identity over time, and none decomposes the estimate into detection versus association. Second, every one of these methods still *runs the model on the target* and inspects its predictions, confidences, or agreement — it estimates the score of an *already-executed* model. We instead forecast from properties knowable *before the model is ever run* on the target.

2.4 Distribution shift in camera-trap and wildlife vision

That camera-trap models degrade at new locations is well established. *Terra Incognita* [2] showed that recognisers excellent at trained camera sites fail sharply at unseen ones; WILDS [8] consolidated this (via iWildCam) as a canonical real-world shift; and PanAf-FGBG [3] demonstrated that behaviour-correlated *backgrounds* causally drive the drop, proposing a latent-space mitigation. These works are the empirical backdrop for our environment axis — but they share two features that mark the gap. They **measure or mitigate** degradation after the fact rather than *predicting* per-site or per-species reliability in advance, and they study **classifiers or behaviour recognisers**, not promptable trackers, scored by a single accuracy with no detection/association split. Most tellingly, a recent unified study of camera-trap recognition over time [7] explicitly names “predicting when zero-shot models will succeed at a new site” as an open question — and stops at naming it. This dissertation builds the predictor that question asks for.

2.5 What should predict transfer

Our candidate predictors are motivated by representation-learning results. Vision foundation models encode taxonomic structure across the tree of life [13], which motivates a *taxonomic* distance and, because appearance and phylogeny are correlated, a *visual* distance; we compute the latter from mask-cropped DINOv2 [11] embeddings, with CLIP [12] as a robustness variant. The camera-trap location results above [2, 3] motivate an *environment* distance from the scene background, and per-video timestamps motivate a *temporal* gap. Crucially, all four are computable without any label of the target animal, and all are measured against a frozen reference set, so no target information can leak into a predictor.

2.6 The SA-FARI benchmark

SA-FARI [15] is a large open multi-animal camera-trap tracking dataset (11,609 densely annotated videos spanning 99 species with a seven-level taxonomy, 741 locations on four continents, and per-video timestamps), released with SAM3 and scored by the official VEval evaluator. It ships with abundant *hard negatives* — prompts for absent species that must return nothing — which we keep, as they drive the false-positive side of detection. Two properties of the shipped split shape our design and merit a precise statement. Inspection of the released annotations

shows the train/test split is disjoint by **location** (test camera sites are unseen) but that its *species are largely shared* across the split; a genuine “unseen species” axis is therefore not directly available from the official split. Because SAM 3 is frozen and zero-shot on *all* of SA-FARI, however, the split reflects nothing the model learned — it is only our choice of a *reference distribution* against which distances are measured. This licenses a species-disjoint *reference/probe partition* of the same videos for the primary (species-novelty) experiment, complemented by the official location split for the environment experiment; the two are near-orthogonal probes of the same data, and their construction is detailed in Chapter 3.

2.7 Summary and gap

Predictable OOD generalisation is established for classifiers scored by a scalar [1, 5, 6, 10] and, very recently, for static detectors and segmenters [14, 16]; camera-trap vision has thoroughly documented, but not forecast, location-driven degradation [2, 3, 7, 8]. No prior work predicts, *before running the model* and from label-free distances, the zero-shot transfer of a **promptable video tracker**, and none separates that prediction into **detection** and **association**. Asking that question rigorously — an out-of-sample test, over SA-FARI’s species-by-place-by-time structure, of whether label-free distances predict SAM 3’s transfer, decomposed into **pDetA** and **pAssA** — is the contribution of this dissertation. As Chapter 4 reports, the honest answer is negative: transfer is not distance-driven.

Chapter 3

Data and Methodology

3.1 Research design and evaluation criteria

This work is a predictive-modelling study: we ask whether label-free *distances* measurable in advance (X) predict a frozen tracker’s zero-shot performance ($Y = \text{pDetA}/\text{pAssA}$), and which factors govern that transfer for *finding* versus *following* an animal. Because the target is a bounded score and the observations are grouped by species and location, the analysis is a weighted regression with group-aware, out-of-sample validation rather than a tuned tracker.

Definition of success and the decision rule. The intended deliverable is a *validated, out-of-sample* predictor of $\text{pDetA}/\text{pAssA}$ from label-free distances, with honest confidence intervals. A distance is taken to predict transfer only if it satisfies **both** criteria: (i) its coefficient’s group-cluster-bootstrap 95% confidence interval excludes zero, *and* (ii) the grouped cross-validated model beats a mean-predictor baseline by a margin that is significant under a paired group-bootstrap. **H1** (detection $\leftarrow$ species novelty) and **H2** (association $\leftarrow$ environment difficulty) are supported only when their respective distances meet this bar on the split that isolates them; if neither does under these deliberately powerful tests, we conclude **H0** — distance explains little variance — which is itself a valid result and motivates the representational-probing direction of Chapter 5.

3.2 Dataset, splits, and the analysis unit

We use SA-FARI [15] throughout. In the released `_ext` annotations, a probe is a (video, text-prompt) pair; the animal’s species and its seven-level taxonomy are resolved through the prompt’s `category_id`, while per-video `location_id` and creation timestamp give place and time. We aggregate scores to a *cell* — a (species, location, time) group — and attach support counts (n_{frames} , n_{masklets} , n_{videos}) used for weighting and as covariates. Hard negatives (prompts for absent species) are kept, as they drive the false-positive side of detection.

Two complementary splits. Inspection of the annotations shows the shipped split is disjoint by *location* but shares species across train and test (Section 2.6); on that split species-distance is ≈ 0 for every cell, so it cannot test species novelty. Because SAM3 is frozen and zero-shot on all of SA-FARI, the split only defines our *reference distribution*, so we are free to repartition the same videos. We therefore use two near-orthogonal experiments and run inference only once (scoring is per-video and split-agnostic):

Split A — species hold-out (primary).

Pooled species are partitioned by a *leave-one-species-out* view, so a held-out species’ distance is to the nearest *other* species; species-distance varies while the environment stays largely familiar, isolating H1. Validated by leave-species-out cross-validation.

Split B — location hold-out (secondary).

The official train/test split, whose unseen locations isolate H2 and whose benchmark comparability carries the Phase-1 score-sanity check.

Split A is a *constructed* hold-out — legitimate because the model is frozen, and reported as such.

3.3 Measuring transfer

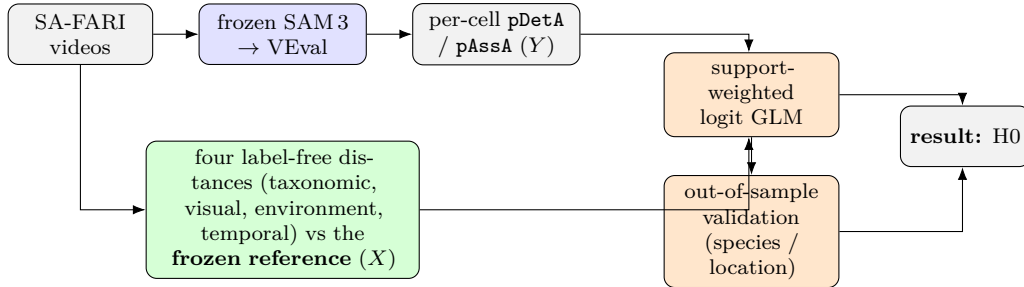


Figure 3.1: The pipeline. *Top:* frozen SAM3 is run over the videos and scored by the official evaluator into per-cell detection/association scores (the target Y). *Bottom:* four label-free distances are computed from the videos and ground-truth masks against a frozen reference set (the predictors X). A support-weighted logit GLM relates X to Y and is validated out of sample by holding out whole species and whole locations; SAM3 is never trained.

SAM3 [4] is run frozen over the probe videos (species-specific prompts as the primary condition, a generic “animal” prompt for robustness) and scored with the official VEval evaluator — the metric is never re-implemented — giving per-cell pDetA/pAssA/pHOTA [9] with their support. A single inference pass over the union of Split A and Split B probes serves both experiments; predictions are stored per (video, prompt) so that the several species-prompts issued on one video do not collide.

No retraining; validity of the frozen protocol. It bears stating plainly, as it is easily misread: **SAM3 is never retrained or fine-tuned in this work.** The single object we fit is the small support-weighted logistic regression that maps the label-free distances X onto the observed scores Y ; the tracker itself is the public checkpoint, run once, its weights untouched. Reusing that one inference across both splits is therefore sound rather than a leak: because the model is frozen, a cell’s score does not depend on which reference partition we later overlay to compute its distances. This also fixes the benchmark’s status here. SA-FARI is a held-out, *zero-shot* benchmark for the released SAM3: the SAM3 report [4] does not list SA-FARI among its training sources, and the SA-FARI paper [15] reports the released checkpoint zero-shot (the ≈ 0.65 mask-pHOTA ballpark we reproduce at Gate 1) *alongside* separately fine-tuned rows whose gain we deliberately forgo. We can therefore verify non-membership at the *dataset* level; we cannot verify it frame-by-frame, since SAM3’s full pretraining corpus is undisclosed. We do not claim otherwise: every distance is measured against the SA-FARI *train reference*, not against SAM3’s true (unknown) pretraining distribution, and the familiarity proxy (Section 3.5, planned) is the hedge for exactly this gap.

3.4 The frozen reference (leakage firewall)

For each split the reference (“seen”) species, locations, and taxonomy are frozen once; every distance is then computed against that reference only, so no probe information can enter a feature — verified by a unit test that swaps the reference and checks the distances move as

expected. We assert the axis that is genuinely disjoint (location for Split B; the held-out species for Split A) and *report* any residual species overlap honestly rather than asserting it away.

3.5 Label-free distance features

All distances are label-free for the target and computed against the frozen reference.

3.5.1 Taxonomic distance

Lowest-common-ancestor tree steps from the probe’s species to the nearest reference species (shared genus = 1, family = 2, order = 3, ...); the leave-species-out reference keeps this non-trivial for shared species.

3.5.2 Visual distance

Cosine distance between DINOv2 [11] embeddings of the *ground-truth-mask-cropped* animal and the nearest reference-species prototype, with a CLIP [12] encoder as a robustness check. Using the dataset’s ground-truth masks means this feature needs no SAM 3.

3.5.3 Environment distance

Embedding distance between the animal-masked-out background scene and the nearest reference location, plus an interpretable day/night–infrared flag and a clutter score derived from colour statistics.



(a) frame + ground-truth mask

(b) *visual*: mask-cropped animal

(c) *environment*: animal erased

Figure 3.2: The two image-based distances on a real camera-trap frame (a collared peccary). **(a)** the frame with its ground-truth mask (orange); **(b)** the mask-cropped animal that the *visual* distance fingerprints (background zeroed); **(c)** the same frame with the animal mean-inpainted, which the *environment* distance fingerprints. The visible timestamp drives the *temporal* gap. None of these features runs SAM 3.

3.5.4 Temporal gap

The minimum absolute year gap between a cell’s footage and the nearest reference footage.

3.5.5 Animal size (confound control)

Mean log ground-truth mask area per species, included as a nuisance covariate to separate genuine visual novelty from the fact that visually-distinctive species tend to be large and thus easier to segment.

3.5.6 SAM 3 familiarity proxy

Separability of the target species in SAM 3’s own feature space — a hedge against the unknown true pretraining set. (*Planned robustness check; see Limitations.*)

3.6 Statistical modelling and hypothesis testing

3.6.1 Per-target regression

Each bounded target (pDetA, pAssA) is modelled *separately* with a support-weighted logit-link (fractional-logit) regression over the standardised distances plus the scene covariates and a $\log(n_{\text{frames}})$ covariate, so that “rare” is never mistaken for “far”. Detection and association are analysed apart because their decomposition is the intellectual core of the project.

3.6.2 Honest, group-aware inference

Naive model confidence intervals are invalid here for two reasons: support-weighting by frame count inflates the effective sample size, and the distances are constant within a species or a location, so the hundreds of cells are really only tens of independent groups (pseudo-replication). We therefore report *group-cluster-bootstrap* intervals — resampling whole species and whole locations, refitting, and taking percentile intervals — as the honest confidence interval for every coefficient.

3.6.3 Out-of-sample validation and the significance test

The fitted models are validated *out of sample* by holding out whole groups (leave-species-out on Split A, leave-location-out on Split B) and predicting the held-out group. The out-of-sample gain over a mean-predictor baseline is tested with a *paired group-bootstrap* that resamples whole held-out groups, yielding a confidence interval and a one-sided *p*-value on the gain — the criterion (ii) of Section 3.1.

3.6.4 Variance attribution and confound control

Variance is attributed to each factor by dominance/commonality analysis with a VIF collinearity check — not by four univariate fits — and raw correlations are reported as a model-free cross-check. Where a distance reaches significance, we refit with the animal-size covariate to test whether the effect is a size confound rather than novelty.

3.6.5 The intrinsic-difficulty pivot and the nested decomposition

Because the sole apparent effect proved to be a size confound, we ask a complementary question: is transfer governed by the intrinsic *difficulty* of the target imagery rather than its *novelty*? We promote the label-free scene properties already measured for the environment axis — low-light/IR (`achromatic_fraction`), clutter, and the night/IR flag — from nuisance covariates to predictors of interest, and test them against the same out-of-sample bar. To separate a genuine difficulty effect from the size confound, we fit a *nested* sequence of models — support only; size only; difficulty only (no size); difficulty plus size — and compare their out-of-sample gains, so that any apparent difficulty effect that is really size is exposed as such (Chapter 4, Table 4.4). We also report each scene property’s *cell-level* correlation alongside its location-level value, because aggregation inflates correlations (the ecological-correlation fallacy).

Detecting a signal, not asserting its absence. These tests are demonstrably able to detect a real effect: the animal-size covariate not only has an interval excluding zero but *validates out of sample* at the same whole-group hold-out bar the distances fail (Chapter 4), and the estimation and cross-validation machinery is unit-tested to recover planted effects on synthetic data. A null under a test that fires for size is therefore evidence of *absence of signal*, not of *absence of power*.

3.7 Detection precision: false positives on hard negatives

The `pDetA` model conditions on the animal being present, so it measures recall-given-present, not hallucination. On hard negatives (the queried species is absent) a correct run returns no masklet; we read the prediction directly, flag any returned masklet above a score threshold as a false positive, attach each species’ taxonomic and visual distance, and test — at the species level, to avoid pseudo-replication — whether the hallucination rate rises with novelty.

3.8 Reproducibility, implementation, and constraints

The model is frozen and scored by the official evaluator; the seen set is frozen before any distance is computed; hard negatives are retained; and all inference is single-GPU and inference-only. The pipeline is config-driven (every constraint is a toggle, so an ablation flips config, not code) and covered by hermetic tests; every reported number is regenerated from the repository, and split definitions, seeds, and reference sets are persisted so both experiments are reproducible end to end.

Chapter 4

Results and Evaluation

4.1 Measurement

Zero-shot SAM 3 on the SA-FARI test split reproduces the published ballpark (Gate 1); the support-weighted per-cell aggregate is in Table 4.1.

Table 4.1: Zero-shot SAM 3 on the SA-FARI test split — support-weighted mean over 346 scored cells (of 1447).

Metric (mask)	Value
pDetA	0.527
pAssA	0.546
pHOTA	0.533

4.2 The fitted law

Table 4.2 gives the standardised logit-GLM coefficients for detection and association.

Table 4.2: Standardised logit-GLM coefficients with *group-cluster-bootstrap* 95% CIs (resampling whole species / locations, so the interval respects pseudo-replication and the support weighting), for detection (pDetA) and association (pAssA). Every distance’s CI spans zero.

Factor	pDetA coef. [CI]	pAssA coef. [CI]
Taxonomic distance	-0.362 [-3.87, 0.09]	-0.368 [-3.40, 0.08]
Visual distance	-0.144 [-0.50, 0.14]	-0.122 [-0.49, 0.18]
Environment distance	0.042 [-0.19, 0.24]	0.013 [-0.23, 0.21]
Clutter	-0.066 [-0.38, 0.29]	-0.087 [-0.41, 0.28]
Night / IR	-0.405 [-0.94, 0.11]	-0.462 [-1.02, 0.11]
log(support)	-0.216 [-0.51, 0.17]	-0.224 [-0.53, 0.18]

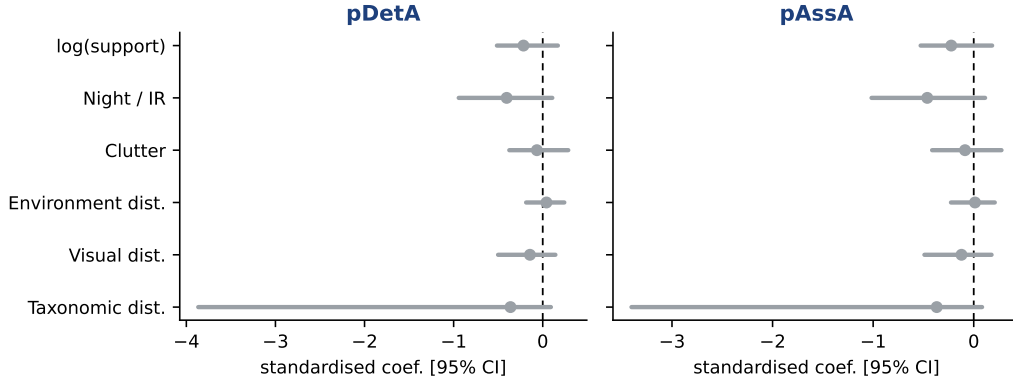


Figure 4.1: Standardised GLM coefficients with group-cluster-bootstrap 95% CIs, for detection (pDetA) and association (pAssA). *Every distance’s interval crosses zero (grey) — none is distinguishable from no effect.*

4.3 Variance partitioning and the intrinsic-difficulty decomposition

A dominance (LMG/Shapley) analysis confirms the distances carry almost nothing: together the four explain only $\approx 3\%$ of the variance in detection, no single factor dominates, and every VIF is ≈ 1 (Table 4.3) — so the null is not an artefact of collinearity masking a real effect. Because the

Table 4.3: Variance partition for detection (pDetA): each distance’s unique share of R^2 (LMG/Shapley) and its VIF. The four distances together explain only $\approx 3\%$ of the variance, no factor dominates, and every VIF ≈ 1 — the null is not masked by collinearity.

Factor	unique R^2	share	VIF
Taxonomic	0.014	48%	1.06
Visual	0.012	39%	1.06
Environment	0.004	13%	1.00
Temporal	0.000	0%	–

one apparent effect (visual distance, below) turned out to be an animal-size confound, we pivot the hypothesis from *novelty* to intrinsic *difficulty*: we promote the label-free scene properties — low-light/IR (`achromatic_fraction`), clutter, and the night/IR flag, each computed per location from the frames with no SAM3 run and no target label — from nuisance covariates to predictors of interest, and test them at the same bar. Table 4.4 decomposes the outcome. Taken *with* an animal-size covariate each difficulty signal *appears* to validate (leave-species-out and leave-location-out $\Delta\text{MAE} \approx +0.017$, both significant), but a nested decomposition shows the gain is entirely the size covariate: log-area *alone* carries it ($\Delta\text{MAE} = +0.017$, $p = 0.001/0.014$), while low-light *alone* does *not* validate ($p = 0.12/0.23$) and adds only $\approx +0.002$ over size. The low-light effect is also far weaker than it first looks: the location-level night/IR–pDetA correlation of $r = -0.377$ is an aggregation artefact (the ecological-correlation fallacy) — at the cell level it is only -0.13 , and clutter’s cell-level correlation is 0.00 . The intrinsic-difficulty pivot is therefore **also null**: the only label-free property that predicts detection is animal size.

Table 4.4: Nested size decomposition of the intrinsic-difficulty test (detection, 346 cells). Each entry is the out-of-sample MAE gain over a mean-predictor baseline, Δ [95% CI] (p), under leave-species-out and leave-location-out. The entire gain sits in the size covariate (log-area); low-light *alone* does not validate — the difficulty signals only pass the bar by riding on size.

Model	Leave-species-out Δ [CI] (p)	Leave-location-out Δ [CI] (p)
log(support) only	+0.001 [-0.007, +0.008] (0.432)	+0.002 [-0.005, +0.008] (0.291)
log(area) only [<i>size</i>]	+0.015 [+0.002, +0.028] (0.014)	+0.017 [+0.009, +0.025] (0.001)
Low-light only (no size)	+0.003 [-0.006, +0.011] (0.233)	+0.004 [-0.003, +0.011] (0.123)
Low-light + size	+0.017 [+0.004, +0.032] (0.011)	+0.020 [+0.011, +0.030] (0.000)

4.4 Out-of-sample validation

Table 4.5 reports whole-group hold-out error against a mean-predictor baseline. This is the headline result and it is a **null**: on both the location and the species schemes, and for both targets, no distance model beats simply predicting the mean — every Δ MAE is within noise of zero and no gain is significant under the paired group-bootstrap. **This is not an underpowered test.** The same GLM and grouped cross-validation, on the same cells, *do* detect a real signal when one exists: animal size (log-area) validates out of sample at the identical bar (Δ MAE = +0.017, $p < 0.017$ on both schemes; Table 4.4). A test that fires for size yet stays flat for the distances is evidence of a genuine *absence* of before-running label-free signal, not of low power.

Table 4.5: Out-of-sample error: whole-group hold-out (leave-species-out / leave-location-out) vs a mean predictor. Δ is the MAE gain over baseline; its bracketed 95% CI and one-sided p come from a paired bootstrap resampling whole held-out groups. No hold-out’s gain is significant.

Hold-out	Target	n	MAE	Baseline	Δ [95% CI]	p
location	pAssA	346	0.289	0.295	+0.007 [-0.005, +0.018]	0.13
location	pDetA	346	0.293	0.298	+0.004 [-0.007, +0.015]	0.23
species	pAssA	346	0.289	0.295	+0.006 [-0.006, +0.017]	0.16
species	pDetA	346	0.293	0.298	+0.004 [-0.006, +0.015]	0.24

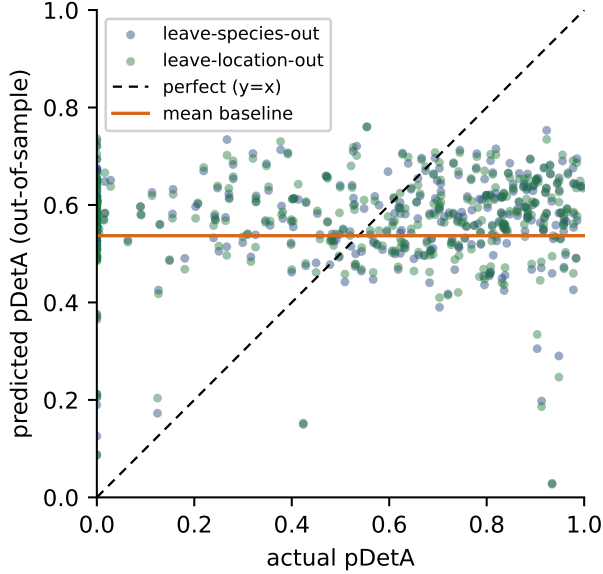


Figure 4.2: Out-of-sample predictions vs the truth for detection (pDetA), on held-out species and locations. The predictions collapse onto the mean baseline (orange) instead of tracking the diagonal (dashed) — the distances carry no out-of-sample predictive power.

4.5 Species hold-out (H1): novelty and the size confound

The primary novelty test — leave-one-species-out over the pooled train+test species (99 species, 1025 scored cells) — is in Tables 4.6–4.8. Out-of-sample prediction is not significant on any hold-out (Table 4.8). The only coefficient whose CI excludes zero, *visual distance* (Table 4.7), is positive — the wrong sign for H1 — and is a **confound**: visually-distinctive species are larger (visual $\leftrightarrow$ log-area $r = 0.22$) and larger animals score higher (log-area $\leftrightarrow$ pDetA $r = 0.30$, the strongest raw correlate of the score). Adding an animal-size covariate shrinks the visual coefficient from +0.37 to +0.22 (CI now spanning zero) while size itself is the significant driver, so “visual novelty helps” was really “bigger animals are easier”. Taxonomic novelty is flat ($r \approx 0.00$). Novelty does not predict transfer.

Table 4.6: Zero-shot SAM3 on the pooled train+test set — support-weighted mean over 1025 scored cells (of 2126).

Metric (mask)	Value
pDetA	0.536
pAssA	0.597
pHOTA	0.559

Table 4.7: Standardised logit-GLM coefficients with *group-cluster-bootstrap* 95% CIs (resampling whole species / locations, so the interval respects pseudo-replication and the support weighting), for detection (pDetA) and association (pAssA).

Factor	pDetA coef. [CI]	pAssA coef. [CI]
Taxonomic distance	0.202 [-0.09, 0.43]	0.209 [-0.06, 0.42]
Visual distance	0.369 [0.11, 0.67]	0.271 [0.04, 0.54]
Environment distance	0.043 [-0.15, 0.17]	0.023 [-0.15, 0.13]
Clutter	-0.284 [-0.52, 0.02]	-0.230 [-0.46, 0.06]
Night / IR	-0.286 [-0.77, 0.22]	-0.439 [-0.92, 0.06]
log(support)	-0.020 [-0.14, 0.16]	0.051 [-0.06, 0.23]

Table 4.8: Out-of-sample error: whole-group hold-out (leave-species-out / leave-location-out) vs a mean predictor. Δ is the MAE gain over baseline; its bracketed 95% CI and one-sided p come from a paired bootstrap resampling whole held-out groups. No hold-out’s gain is significant.

Hold-out	Target	n	MAE	Baseline	Δ [95% CI]	p
location	pAssA	1025	0.270	0.270	-0.000 [-0.006, +0.005]	0.52
location	pDetA	1025	0.274	0.274	-0.001 [-0.007, +0.006]	0.58
species	pAssA	1025	0.273	0.270	-0.003 [-0.014, +0.008]	0.74
species	pDetA	1025	0.277	0.274	-0.004 [-0.015, +0.009]	0.74

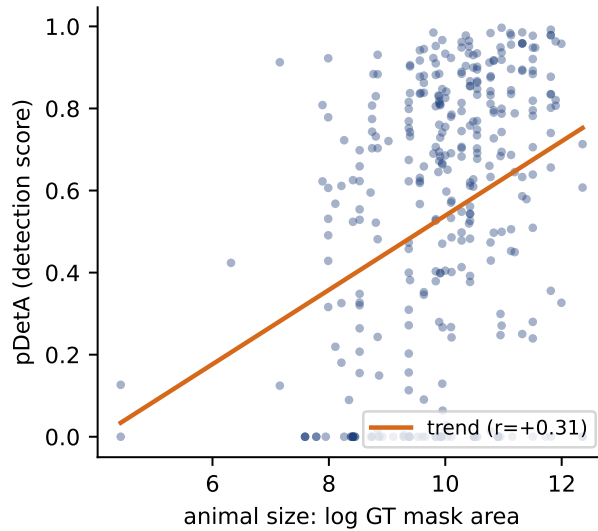


Figure 4.3: The one label-free property that predicts detection: animal size. Per-cell pDetA rises with the log ground-truth mask area ($r = +0.31$) — the confound behind the apparent visual-distance effect.

4.6 False positives / hallucination

The detection score conditions on the animal being present, so it measures recall, not hallucination. The 1486 *hard negatives* (queries for absent species) ask the complementary question, and are the closest the study comes to a validated before-running signal. SAM3 returns a masklet anyway about **9.8%** of the time. This hallucination rate does *not* rise with taxonomic novelty ($r \approx +0.06$, n.s.); it *falls* with visual distinctiveness ($r \approx -0.33$), and this survives an animal-size control (partial -0.37) — a genuine, size-independent, label-free correlate, though its direction (“distinctive = safer”) is the *opposite* of “novelty hurts”. Crucially it is a **correlate, not a predictor**: predicting a held-out species’ hallucination rate from visual distance beats a mean baseline by only $+0.008$ MAE and *just misses* out-of-sample significance ($p = 0.053$; Table 4.9). Even the study’s best signal does not clear the bar.

Table 4.9: False positives on the 1486 hard negatives (113 species): the overall hallucination rate is 9.8%. More visually distinctive species hallucinate *less* — a real, size-independent correlate — but it does not validate out of sample (leave-species-out $p = 0.053$), so it is a correlate, not a predictor.

Test (species-level)	statistic	95% CI	verdict
FP rate vs taxonomic distance	+0.06	$[-0.17, +0.23]$	n.s.
FP rate vs visual distance	−0.33	$[-0.43, −0.23]$	significant
visual, size-controlled (partial)	−0.37	$[-0.47, −0.28]$	significant
FP rate vs animal size	+0.13	$[-0.06, +0.31]$	n.s.
leave-species-out OOS gain	+0.008	$[-0.002, +0.017]$	$p = 0.053$

4.7 Ablations and robustness

Factor ablation. Refitting detection with each distance dropped in turn barely moves the fit or the (null) out-of-sample gain (Table 4.10): no single distance carries signal. The only design that changes anything is adding the animal-size covariate, which alone lifts the out-of-sample gain to significance ($\Delta\text{MAE} \approx +0.02$, $p \leq 0.02$) — again consistent with size being the sole real correlate. Removing the support covariate does not rescue a distance effect either. The variance partition (Table 4.3) tells the same story from the other direction.

Table 4.10: Factor ablation for detection (pDetA): McFadden pseudo- R^2 and the out-of-sample MAE gain over a mean baseline under leave-species-out (LSO) and leave-location-out (LLO). Dropping any single distance barely changes the fit or the (null) out-of-sample gain — no factor carries signal; only adding the animal-size covariate moves the score.

Design	pseudo- R^2	LSO Δ (p)	LLO Δ (p)
full (4 distances)	0.055	+0.004 ($p=0.24$)	+0.004 ($p=0.23$)
drop taxonomic	0.039	+0.000 ($p=0.47$)	+0.000 ($p=0.49$)
drop temporal	0.055	+0.004 ($p=0.24$)	+0.004 ($p=0.23$)
drop visual	0.048	+0.004 ($p=0.22$)	+0.004 ($p=0.28$)
drop environment	0.055	+0.006 ($p=0.15$)	+0.005 ($p=0.17$)
+ animal size	0.119	+0.018 ($p=0.02$)	+0.020 ($p=0.00$)
− support covariate	0.035	−0.002 ($p=0.72$)	−0.003 ($p=0.73$)

Design robustness (future work). Three further checks — a CLIP rather than DINOv2 visual encoder, whole-frame rather than mask-cropped visual embeddings, and a generic “animal”

rather than a species-specific prompt — each require a fresh GPU inference or embedding pass and are left for future work. Given that the null is flat across the factor ablation, the variance partition, both hold-out schemes, and the intrinsic-difficulty pivot, a change of encoder, crop, or prompt is unlikely to overturn it.

Chapter 5

Discussion

5.1 Interpretation

Neither hypothesis is supported. H1 (detection falls with species novelty) and H2 (association falls with environment difficulty) both fail the pre-registered bar: no label-free distance beats a mean predictor out of sample, and the one coefficient that reached significance — visual distance, with the wrong sign — is an animal-size artefact. The follow-up intrinsic-difficulty pivot fares no better: low-light, clutter and night/IR predict detection only by proxying animal size, which alone carries the out-of-sample gain. So the honest reading is H0: transfer is *not* distance-driven, and the only label-free property that predicts detection is object size — the very quantity introduced as a nuisance control.

The detection–association decomposition, the intellectual core of the design, has a second, quieter finding: association is a near-degenerate target on this data. Because most cells contain a single object, **pAssA** tracks **pDetA** almost exactly, leaving little identity-specific variance for any feature to predict. The decomposition is therefore honestly a detection story; we report the association null with those divergence figures as its explanation rather than as a separate failed search.

5.2 Relation to prior label-free evaluation

The null sits against a literature of *positive* results, and two axes separate every one of those results from ours — which is why the disagreement is only apparent. The first axis is the **signal**. Accuracy- and Agreement-on-the-Line [1, 10], ATC [6] and AutoEval [5] all read the model’s *own behaviour on the target* — its confidence, its agreement across a model pair, or (for AutoEval) a feature-distribution distance computed after passing the target images through the network. Those are *after-running* estimators. We instead forecast from *external*, label-free distances known *before* a single frame is processed — the strictly harder, and for a practitioner deciding whether to spend compute, the more useful regime. The second axis is the **output**. Every predecessor targets a *scalar* classifier accuracy [1, 5, 6, 10] or the quality of a *static* detector/segmenter [14, 16]; none concerns identity maintained over time, and none is decomposed. Ours is the first study to ask the question of a promptable *video tracker*, split into detection (**pDetA**) and association (**pAssA**). AutoEval [5] is the closest methodological cousin — it, too, regresses a performance score on a distributional feature distance — and it is precisely the recipe whose strong classifier-level correlation does *not* reproduce once the distance is measured before running and the target is a tracker. Our result therefore does not contradict these positives; it extends the question to a regime where the label-free-distance shortcut stops working, and reports honestly that it does.

5.3 Why label-free distances do not yield a deployment estimator

The field problem that motivated this work — telling a practitioner, before spending compute, whether SAM3 is trustworthy for a new (species, place) — is not solved by label-free distances. That is a negative but useful answer: the intuition that “far-from-training implies unreliable” does not hold for a promptable video tracker, so a deployment gate cannot be built from taxonomic, visual, environmental or temporal distance to a reference set alone. The one predictive label-free signal, animal size, is too coarse and too obvious to serve as such a gate. What the negative result does establish is a rigorous baseline: any future before-running estimator must clear the same whole-group hold-out bar that these well-motivated distances failed.

5.4 Limitations and threats to validity

Three limitations bound the claim. First, distances are measured against the SA-FARI train split, *not* SAM3’s true pretraining corpus (Meta’s SA-Co), which is undisclosed; the null is scoped to distance-from-SA-FARI, and a coverage-anchored proxy for the real corpus remains untested (the deferred familiarity proxy is one such hedge). Second, the analysis is a single dataset and a single frozen backbone, and — since one inference pass is reused across both splits — the species and location experiments are two overlays on the *same* scored cells, not independent replications; they give near-orthogonal but not statistically independent evidence. A genuinely independent test of the null therefore requires re-running the whole pipeline on a *different* tracking dataset (for example a mask-native subset of BURST, or the camera-trap MammAlps), which we leave to future work. Third, the effective sample is tens of independent species/locations, not hundreds of cells — which is why inference is group-clustered throughout, and why aggregate correlations (e.g. the location-level night/IR effect) are reported alongside their much weaker cell-level values to avoid the ecological-correlation fallacy. The open branch this work does not pursue is representational probing — whether SAM3’s own features encode the species.

5.5 Ethical considerations

The data are anonymised camera-trap footage released under a non-commercial licence; location identifiers are already anonymised in SA-FARI. The negative result carries a responsible-deployment message: because label-free distance does *not* reliably flag when SAM3 will fail, conservation teams should not treat low-distance-from-training as a safety guarantee, and reliability should be checked with labelled spot-audits rather than assumed.

Chapter 6

Conclusion

6.1 Summary of findings

We asked whether SAM 3’s zero-shot promptable-tracking accuracy on an unseen (species, place) can be predicted, before running the model and without target labels, from four label-free distances — separately for detection and association. Under a deliberately powerful out-of-sample test the answer is no: no distance beats a mean-predictor baseline on either a species or a location hold-out, the one apparent effect is an animal-size confound, and a follow-up intrinsic-difficulty pivot collapses to the same size effect. Only object size predicts detection, modestly, and it is the nuisance the analysis controls for. Association carries almost no signal distinct from detection. The result is a rigorous, decomposed null.

6.2 Contributions

To our knowledge this is the first study to test, out of sample, whether a *promptable video tracker’s* zero-shot transfer is predictable from label-free, before-running signals, and the first to decompose that question into detection and association. Its contribution is the honest negative answer and its explanation: transfer is not distance-driven; the apparent signals reduce to an animal-size confound; and the test is shown to detect that one real signal when present, so the null reflects an absence of signal, not of power.

6.3 Future work

Two directions remain open. First, representational probing — whether the target species is separable in SAM 3’s own feature space — as a hedge against the unknown pretraining corpus. Second, broadening beyond a single dataset and backbone, and to targets with richer association structure (crowded, multi-object scenes) where the detection–association decomposition would have more to distinguish.

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Appendix A

Mathematical definitions

This appendix collects the exact definitions used in Chapter 3. The tracking metric is computed by the official evaluator and never re-implemented; the distances and the statistical procedures are ours and are stated here as they appear in the code.

A.1 The promptable-tracking metric

Promptable HOTA decomposes, at each localisation threshold α , into a detection and an association term [9]:

$$\text{HOTA}_\alpha = \sqrt{\text{DetA}_\alpha \cdot \text{AssA}_\alpha}, \quad \text{pHOTA} = \frac{1}{|\mathcal{A}|} \sum_{\alpha \in \mathcal{A}} \text{HOTA}_\alpha, \quad (\text{A.1})$$

with $\mathcal{A} = \{0.05, 0.10, \dots, 0.95\}$. We analyse the two components separately as the targets $\text{pDetA} = \text{DetA}$ (did it *find* the animals?) and $\text{pAssA} = \text{AssA}$ (did it *keep* each identity?), both bounded in $[0, 1]$. Scores are computed per cell by the vendored **VEval** evaluator.

A.2 Label-free distances

Let the frozen reference be the “seen” set of a split; every distance below is computed against that reference only.

Taxonomic distance. Write each species’ seven-level path $\pi(s) = (\text{kingdom}, \dots, \text{species})$, defined only for species with a complete taxonomy. Let $\text{shared}(a, b)$ be the length of the longest common *non-empty* prefix of two paths. The tree-step distance and the nearest-reference distance are

$$\tau(a, b) = L - \text{shared}(a, b) \quad (L = 7), \quad d_{\text{tax}}(s) = \min_{s' \in \mathcal{R}, s' \neq s} \tau(\pi(s), \pi(s')), \quad (\text{A.2})$$

where the exclusion $s' \neq s$ is the leave-one-species-out rule (Split A). Shared genus gives $\tau = 1$, family 2, order 3, fully disjoint 7; species without a full taxonomy take $d_{\text{tax}} = \text{NaN}$.

Prototype and nearest cosine distance. Given a set of ℓ_2 -normalised embeddings $\{v_i\}_{i=1}^N$, the prototype and the distance from a query v to a set of prototypes $\{p_j\}$ (optionally excluding an index e) are

$$p = \frac{\bar{v}}{\max(\|\bar{v}\|_2, 10^{-12})}, \quad \bar{v} = \frac{1}{N} \sum_i v_i, \quad d(v) = 1 - \max_{j \neq e} v^\top p_j. \quad (\text{A.3})$$

Because v and each p_j are unit-norm, $v^\top p_j$ is the cosine similarity, so $d \in [0, 2]$.

Visual and environment distance. Both use DINOv2 embeddings. The *visual* distance embeds the ground-truth-mask-cropped animal (the background is zeroed before cropping), forms one prototype per species, and sets $d_{\text{vis}}(s)$ to the nearest *other* species’ prototype. The *environment* distance embeds the animal-masked-out background (the animal union is replaced by the mean colour of the remaining pixels), forms one prototype per location, and sets $d_{\text{env}}(\ell)$ to the nearest other location. Two interpretable scene covariates accompany the environment distance:

$$\text{achromatic}(f) = \left[\overline{\max_c f_c - \min_c f_c} < \tau_{\text{ir}} \right], \quad \tau_{\text{ir}} = 12, \quad (\text{A.4})$$

$$\text{is_night_ir}(\ell) = \left[\overline{\text{achromatic}(f)} > 0.5 \right], \quad \text{clutter}(\ell) = \overline{\text{Var}(\nabla^2 f)}, \quad (\text{A.5})$$

where the bar denotes a mean over a location’s frames, $\max_c - \min_c$ is the per-pixel channel spread, and ∇^2 is the four-neighbour Laplacian on the greyscale background.

Temporal gap and animal size. For a cell with footage year y and reference years $\mathcal{Y}$, and for the size covariate,

$$d_{\text{temp}} = \min_{r \in \mathcal{Y}} |y - r|, \quad \log_area(s) = \ln(1 + \overline{area(s)}), \quad (\text{A.6})$$

where $\overline{area(s)}$ is the mean ground-truth mask pixel-count over sampled masklets of species s . Size is a *nuisance* covariate, not a hypothesis variable.

A.3 Model and statistics

Support-weighted fractional-logit GLM. Each bounded target $y \in [0, 1]$ is modelled with a Binomial (logit-link) GLM

$$\text{logit } \mathbb{E}[y] = x^\top \beta, \quad \text{fit with variance weights } w = n_{\text{frames}}, \quad (\text{A.7})$$

where the design x standardises each continuous predictor $z = (x - \mu)/\sigma$ (missing values imputed to the training mean μ , zero-variance columns dropped), adds $\log_{\text{support}}$, the standardised $\ln(1 + n_{\text{frames}})$, passes binary `is_night_ir` through, and adds an intercept. Standardisation statistics are learned on training rows only.

Explained deviance and the cross-validation gain. Fit quality is the McFadden pseudo- R^2 , and per-cell out-of-sample gain over a mean predictor is

$$R_{\text{McF}}^2 = 1 - \frac{D}{D_0}, \quad \delta_i = \underbrace{|y_i - \bar{y}|}_{\text{baseline error}} - \underbrace{|y_i - \hat{y}_i|}_{\text{model error}}, \quad (\text{A.8})$$

with D, D_0 the model and null (intercept-only) deviances; $\delta_i > 0$ means the distance model beats “guess the mean” on cell i .

Weighted Pearson correlation. For the species-level false-positive analysis, with weights w (per-species probe counts),

$$r_w = \frac{\sum_i w_i (x_i - \bar{x}_w)(y_i - \bar{y}_w)}{\sqrt{\sum_i w_i (x_i - \bar{x}_w)^2} \sqrt{\sum_i w_i (y_i - \bar{y}_w)^2}}, \quad \bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}. \quad (\text{A.9})$$

Appendix B

Algorithms

Pseudocode for the four non-obvious procedures. The two bootstraps differ deliberately: the cross-validation gain (Alg. 3) is tested *one-sided* (does the model beat the mean?), whereas the false-positive correlation (Alg. 4) is tested *two-sided* (does hallucination change with novelty, in either direction?).

Algorithm 1: Memory-lean, cache-first crop embedding (`embed_crops`).

Input: items (key, record, frame, crop_fn); embedding cache; embedder; chunk size C , workers W
Output: $\{\text{key} \mapsto \text{embedding}\}$

```
1 misses ← [item : cache.get(item.key) = ∅];
2 for each contiguous chunk  $B$  of  $C$  items in misses do
3   group  $B$  by video; for each video do ensure_frames(missing frames);
4   loaded ← map load_frame → crop_fn over  $B$  using  $W$  threads;
5   pending ← [( $k, c$ ) ∈ loaded :  $c \neq \emptyset$ ];
6   if pending ≠ ∅ then
7     vecs ← embedder.embed(crops of pending);
8     for ( $k, \_$ ),  $v$  in pending, vecs do cache.put( $k, v$ );
9   free pending ;                                // bound peak RAM to one chunk
10  every 16 chunks: cache.save() ;                 // resumable checkpoint
11 if misses ≠ ∅ then cache.save();
12 return {item.key ↦ cache.get(item.key)} for all items;
```

Algorithm 2: Group-cluster-bootstrap coefficient CIs (`group_bootstrap_cis`).

Input: scored cells; target; grouping columns $\mathcal{G} = (\text{category_id}, \text{location_id})$; resamples B ; level $\alpha = 0.05$
Output: per-coefficient interval $[\text{ci_lo}, \text{ci_hi}]$

- 1 **foreach** grouping column $g \in \mathcal{G}$ present in the data **do**
- 2 $U \leftarrow$ unique groups of g ; **if** $|U| < 2$ **then** skip;
- 3 **for** $b \leftarrow 1$ to B **do**
- 4 draw $|U|$ groups from U with replacement; sample $\leftarrow$ all rows of the drawn groups;
- 5 refit the GLM on sample; record its coefficient vector ; // skip a singular resample
- 6 $\log \leftarrow$ percentile $\frac{\alpha}{2}$, $\text{hi}_g \leftarrow$ percentile $1 - \frac{\alpha}{2}$, per coefficient;
- 7 **return** conservative envelope $\text{ci_lo} = \min_g \log$, $\text{ci_hi} = \max_g \text{hi}_g$;

Algorithm 3: Paired group-bootstrap cross-validation significance (`oos_predictions + _bootstrap_delta`).

Input: scored cells; target; group column (species or location); resamples B
Output: ΔMAE , its 95% CI, and one-sided p

/* Leave-one-group-out out-of-sample prediction */

- 1 **foreach** held-out group **do**
- 2 refit the GLM on the other groups (own standardisation); predict the held-out cells ;
 // assert train/test groups disjoint
- 3 $\delta_i \leftarrow |y_i - \bar{y}| - |y_i - \hat{y}_i|$ for every predicted cell;
 /* Group bootstrap of the mean gain */
- 4 **for** $b \leftarrow 1$ to B **do**
- 5 draw whole groups with replacement; $\text{boot}_b \leftarrow \text{mean}(\delta \text{ over the drawn cells})$;
- 6 $\text{CI} \leftarrow$ percentiles $[2.5, 97.5]$ of boot ; $p \leftarrow \Pr(\text{boot} \leq 0)$; significant $\Leftrightarrow \text{CI_lo} > 0$;
- 7 **return** $\Delta\text{MAE} = \text{mean}(\delta)$, CI, p ;

Algorithm 4: False-positive / hallucination analysis (`false_positives`).

Input: split; score threshold t ; resamples B
Output: overall FP rate; per-species FP rate vs. novelty with a bootstrap CI

- 1 $\mathcal{H} \leftarrow$ hard-negative probes (queried species absent);
- 2 build a partition whose probe species are the hard-negative species; compute $d_{\text{tax}}, d_{\text{vis}}$ for them;
- 3 **foreach** probe (*video, species*) $\in \mathcal{H}$ with a prediction file **do**
- 4 $\text{fp} \leftarrow [\max_k \text{score}_k \geq t]$; // a masklet returned for an absent species = a hallucination
- 5 attach $d_{\text{tax}}, d_{\text{vis}}$ of the species;
- 6 aggregate to per-species FP rate; $r \leftarrow \text{weighted-Pearson}(\text{FP rate}, \text{distance})$;
- 7 **for** $b \leftarrow 1$ to B **do**
- 8 resample species with replacement; recompute r ;
- 9 $\text{CI} \leftarrow$ percentiles $[2.5, 97.5]$; significant $\Leftrightarrow \text{CI}$ excludes 0 (two-sided);
- 10 **return** overall FP rate, taxonomic-tercile breakdown, r and CI for taxonomic and visual novelty;

Appendix C

Supplementary results